

Night and Day: Diurnal Warming and Structural Transformation in India

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Abstract

This paper finds diverging partial effects of diurnal warming—higher nighttime and daytime temperatures—on agricultural wage-labour shares from decadal Indian Censuses (1981–2011). Though both margins contract grain output and cultivated area; only higher maxima raise harvest prices locally—consistent with a model where warmer nights shock land but warmer days shock land and labour productivity. Warming nights shift seasonal workers and self-cultivators into agricultural labour; warming days push labour to the seasonal margin. Long differences show the labour divergence is rural. In towns, both margins depress non-agricultural worker shares.

Keywords: Climate Change, Structural Transformation, Agriculture, India

JEL Codes: O1, Q54, J43, R11

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Structural transformation—the reallocation of labour from agriculture into non-agricultural employment—is the foundation of economic development (Matsuyama 1992; Herrendorf, Rogerson, and Valentinyi 2014). However, it is unclear whether a warming climate accelerates or arrests the nonagricultural transition. The empirical evidence splits by horizon. In the short run, Colmer (2021) finds that year-to-year temperature shocks reallocate workers out of agriculture in India, and Jessoe, Manning, and Taylor (2017) find that extreme heat depresses formal wage employment, pushing workers to migrate out of rural Mexico.

Yet, the out-migration response present in rural Mexico is largely muted in India, where rural caste-based insurance networks outweigh urban wage gains (Munshi and Rosenzweig 2016). In the long run, Liu, Shamdasani, and Taraz (2023) find that rising temperatures hold labour on the land, increasing agricultural labour shares within Indian districts (1961–2011). Across countries, Nath (2025) shows that despite heat-induced productivity losses, a “food problem” intensifies agricultural employment and specialisation in developing economies. The debate remains unsettled on which way warming temperatures move the “productivity gap” between the agricultural share of output and employment.¹

Prior literature has measured warming undimensionally: through heating or cooling ‘degree days’, a count of extreme days, or binned daily temperatures (Schlenker, Hanemann, and C. 2006; Blakeslee and Fishman 2015; Dell, Jones, and Olken 2014; Somanathan et al. 2021). This paper isolates the ‘level’ effects and the diurnal ‘range’ effects of warming temperatures through a simultaneous specification of nighttime and daytime temperatures. The central result establishes a long-run divergence in agricultural wage-labour shares, rising or falling based on the *time of day* when warming is most prominent. From a decadal panel of Indian Census districts (1981–2011), I find that higher nighttime minimum (T_{min}) temperatures trap agricultural labour, while higher daytime maximum (T_{max}) temperatures release it.

The divergence in agricultural labour shares is rooted in crop and human physiology: crops stand on agricultural land around the clock; farmers are exposed most under daylight. Warmer nights raise nocturnal respiration, spending the carbon that crops fix by day (Peng et al. 2004; Welch et al. 2010); warmer days press field labour against a physiological ceiling, lowering the productivity of labour itself (Dunne, Stouffer, and John 2013; Somanathan et al. 2021). This paper measures decadal-scale adjustments in labour reallocation to rising nighttime and daytime dry-bulb surface-air temperatures—each net of the other and of rainfall, across Indian Census districts, towns, and urban households (NSS) over a harmonised time horizon (1981–2011).

Figure 1 shows the warming geography of Indian districts, by long-differencing the

¹Agriculture still employs roughly 42% of India’s workforce today (World Bank, 2025); this is contemporary motivation, not a figure from the 1981–2011 estimation sample.

average nighttime and daytime temperatures over two endpoints (1981,91 – 2001,11). Within districts, statutory and Census ‘towns’ form the urban component, and villages/gram panchayats aggregate up to the rural component. Although both night and day temperatures ‘levels’ have similar magnitudes of change across districts within their parent administrative zone (the state), the contraction and expansion of their diurnal temperature range (DTR) follows a geographic sorting. Mall et al. (2021) find evidence of asymmetric warming in India beginning 1991. The diurnal temperature range (DTR) contracted in the arid northwest and Indo-Gangetic plains, and expanded in the Deccan peninsula and along the west and east coasts. This setting presents a ‘natural’ experiment to identify local economic effects of nighttime and daytime warming, as a continuous treatment intensity.

This paper establishes three core empirical facts. The central finding is an economically meaningful divergence in agricultural wage-labour shares within districts under diurnal warming — the labour share rises and falls under higher minimum vs maximum temperatures, respectively.

First, I show that warming night and day dry-bulb temperatures negatively impact grain production and land-use within districts; while only daytime temperatures raise their harvest prices locally (Table 1) consistent with the “food problem” (Nath 2025). The effects on area under cultivation are indistinguishable under both shocks ($\beta_{T_{\min}} = -0.22$ and $\beta_{T_{\max}} = -0.22$ log points, $F = 0.01$), failing to reject the null hypothesis of pairwise equality ($\beta_{T_{\min}} = \beta_{T_{\max}}$).² Nighttime warming is suggestive of a lower loss in grain production (-0.14) relative to daytime warming (-0.22), with no detectable change in local harvest prices. Daytime warming acts as a physical supply shock, spiking local harvest prices by 0.13 log points rejecting equality with night warming ($F = 13.34$).

Second, I show that the direction of change in wage-labour shares reverses between night and day warming, simultaneously specified in a district-level panel estimation. I show that a 1°C decadal rise in nighttime minimum temperatures ($\beta_{T_{\min}}$) *increases* agricultural wage-labour shares by +4.4 percentage points within districts, drawing from self-cultivators (-2.8) and seasonal ‘marginal’ workers (-3.3).³ An equivalent 1°C decadal rise in the daytime maximum temperatures ($\beta_{T_{\max}}$) *decreases* agricultural wage-labour shares (-2.6) and raises seasonal worker shares ($+3.2$).

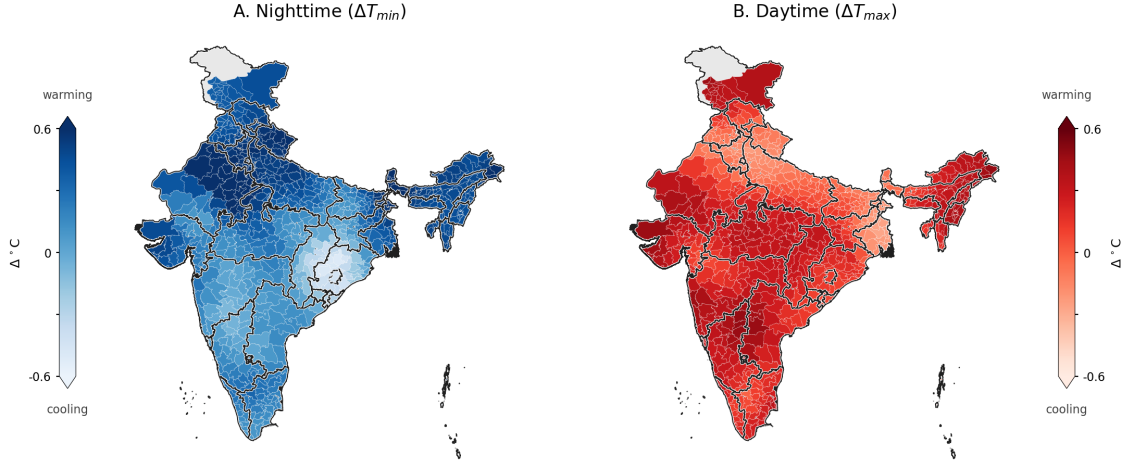
Within districts, the two coefficients reject the null hypothesis of the joint equality ($\beta_{T_{\min}} = \beta_{T_{\max}}$) of their effects on agricultural labour shares ($F = 24.9, p < 0.001$) and seasonal worker shares ($F = 14.7$). The opposing signs are robust to humidity-adjusted wet-bulb temperatures ($+0.086/ - 0.079$), and a long-differenced estimation

²The core grain basket of India is composed of rice, wheat, maize, barley, sorghum, pearl and finger millets.

³The Census of India classifies ‘marginal workers’ as those employed under 183 calendar days preceding the year of enumeration — interpreted as ‘seasonal workers’.

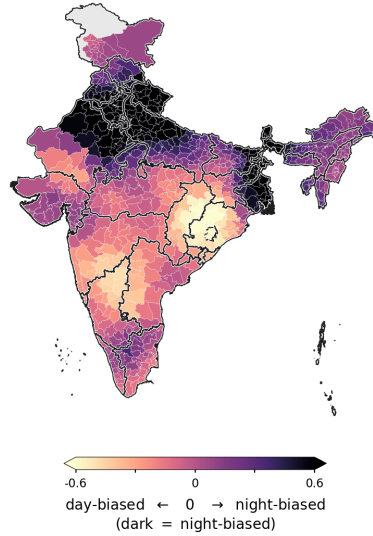
FIGURE 1. The diurnal warming geography of India

Long-differenced change in growing-season temperature, 1981 → 2011



Diurnal warming bias, 1981 → 2011

C. Nighttime – Daytime ($\Delta T_{min} - \Delta T_{max}$)



Notes: Spatial distribution of warming nighttime and daytime temperatures in India. This map plots the change in decadal temperatures over the growing-season (June–Feb), by long-differencing the average between two endpoints: (2001,2011) - (1981,1991). IMD gridded climate to each district centroid by inverse-distance weighting. Panels A and B map the change in the nighttime minimum (ΔT_{min}) and the change in daytime maximum (ΔT_{max})—darker where warming is stronger. Panel C maps the diurnal warming bias ($\Delta T_{min} - \Delta T_{max}$): darker where nights warmed faster (compressing DTR) and lighter where days warmed faster (widening DTR). Across the 639 districts with coverage, nights outpaced days across the Indo-Gangetic north, while days outpaced nights across the peninsular Deccan.

($\beta_{\Delta T_{\min}} = +0.075, \beta_{\Delta T_{\max}} = -0.062$). The divergence is attenuated under a single measure of heat, such as growing-season degree-days ($\beta_{DDs} = +0.036$, online appendix).

Third, the diurnal reversal in signs on wage-labour scales with the relative weight of agriculture in the local economy. Over two-decade long differences, the night-day divergence is strongest among rural workers ($F = 30.9$) and muted among the urban workforce ($F = 1.9$) in districts (Figure 2). Both partitions reflect a secular trend of declining agricultural labour share trends ($\Delta \bar{y}$) between 1981–2011: a $\Delta \bar{y}^{rural} = -0.05$ decline in the rural share and $\Delta \bar{y}^{urban} = -.04$ percentage point decline in the urban share. In towns (Table 3), both measures reverse the declining trend by raising agricultural labour shares, with no detectable night-day divergence ($F = 1.61$). The impacts of warming days ($\beta_{\Delta T_{\max}} = -0.08$) are marginally higher than warming nights ($\beta_{\Delta T_{\min}} = -0.04$) in depressing nonagricultural employment shares within towns ($F = 2.78$).

Over long differences, the district-level night-day rural divergence presents a divide between nighttime warming formalising rural agricultural labour, and daytime warming casualising urban agricultural labour. Together, the two margins jointly converge at the town-level to reduce nonagricultural employment shares, consistent with dampened demand for nonagricultural goods and services (Liu, Shamdasani, and Taraz 2023).

These facts share a common root: nighttime ($T_{\min}$) and daytime warming ($T_{\max}$) reach the farm through different factors of production. A warmer night shocks the land, scaling the productive services of the soil; a warmer day is a neutral productivity loss, eroding the capacity of labour and the underlying land productivity. The online appendix develops a general-equilibrium framework modelling a local agrarian economy operating under partial closures, consistent with the daytime warming supply shock raising harvest prices locally. Section 1 synthesizes these findings under a conceptual framework, discussing assumptions and mechanisms. I additionally test two implications predicted by the framework. The model is formalised mathematically under a “Theoretical Framework” in the online appendix. Additionally, the online appendix presents several robustness decompositions across space, time and measurement; verifying the three central empirical facts are agnostic to specification.

Under a CES agricultural production function where labour and land are gross substitutes (with an elasticity of substitution $\sigma > 1$), the two shocks reach labour demand through different channels. The land-augmenting $T_{\min}$ shock alters the land-to-labour ratio and shifts demand for agricultural labour, independent of its effect on land productivity and yields. The Hicks-neutral $T_{\max}$ shock leaves that ratio intact, contracting the overall scale of production. Therefore, the two move agricultural labour demand in opposite directions. Warmer nights consolidate marginally productive land, shifting cultivation into agricultural wage-labour, fending off the price shock. Warmer days casualise agriculture, transitioning wage labour into seasonal work. Under a unit

elasticity of substitution, the production function is a standard Cobb–Douglas case, where the two shocks are observationally equivalent and insufficient to explain the agricultural labour divergence.

A final piece of suggestive evidence utilizes a near-decadal panel of urban households from three rounds of the National Sample Survey (1987, 1999, 2009) (Ruggles et al. 2024). The daytime heat-displaced labour arrives at the informal, casual end of urban services; corroborating the central mechanism with a parallel data source. Warming nights raise agricultural labour employment shares, warming days raise employment shares in trade, transport and logistics — closely mirroring the decadal pattern of ‘seasonal’ work in urban Censuses. I interpret these as suggestive evidence for future research. The distinction matters for welfare: it separates the nighttime warming shock that halts structural transformation from the daytime warming shock that diverts agricultural wage-employment into a low-productivity, informal churn in urban services.

This paper makes four contributions. First, the diurnal temperature decomposition reconciles conflicting short-and-long run estimates of temperature and structural transformation from Colmer (2021) and Liu, Shamdasani, and Taraz (2023). Warming nights sort labour within agriculture, and warming days reduce formal wage employment. The direction of change observed in agricultural labour shares depends on which end of the day warms. Second, the conceptual framework shows the night/day divergence to be a property of factor bias under gross substitutability of labour and land, carrying an over-identifying restriction—only daytime warming moves the local harvest price ($F = 13.34$) which corroborates partial market integration. Third, I construct a harmonized panel of the Indian economy and climate during an economically important period (1981–2011): covering district level agricultural statistics (ICRISAT), household employment (NSS), and demographics & occupation with a town-level resolution; presenting evidence on local urban economic effects of climate change. Fourth, the paper qualifies the structural-transformation literature (Matsuyama 1992; Herrendorf, Rogerson, and Valentinyi 2014): under sustained warming, towns foreclose entry into nonagricultural employment, arresting structural transformation. Section 1 develops the framework. Section 2 describes the data and the empirical approach. Section 3 reports results. Section 4 concludes.

1. Conceptual Framework

This section lays out the economic logic that maps each temperature margin to the labour responses the rest of the paper estimates. The full general-equilibrium model, with proofs, is in the online appendix.

What the two margins measure. The nighttime minimum and the daytime maximum are the floor and the ceiling of the daily temperature cycle. Holding one fixed, a higher

minimum compresses the diurnal range and a higher maximum widens it, so the decomposition speaks to the range—the quantity agronomy ties to grain filling and heat stress—as much as to the level. The two margins act on different inputs. Crops occupy the field around the clock, so a warmer night works on the standing crop and the land in cultivation, raising respiration and the overnight heat load the crop carries (Peng et al. 2004; Welch et al. 2010). Field labour occupies only the daylight, so a warmer day works on the worker, pressing effort against the physiological ceiling of sustained outdoor exertion and lowering the productivity of labour itself (Dunne, Stouffer, and John 2013; Somanathan et al. 2021). A daily average folds the two channels into one number; the analysis keeps them apart. Equivalently, entering the two margins jointly separates the daily *level* (their mean) from the diurnal *range* (their difference): the level moves the farm’s scale—area, output, and self-cultivation—while the range drives the reallocation between wage labour and the seasonal margin and the harvest price, and a degree-day average retains only the level.

Two shocks, two factors. The framework assigns each margin to the factor it acts on: nighttime warming is a *land-biased* shock that raises or lowers the productive services of cultivated land, while daytime warming is a *Hicks-neutral* loss of total factor productivity. The night margin is distinctive for the factor it lands on rather than the sign of its effect: the nocturnal-respiration and overnight-heat-load channels could in principle either degrade or enhance effective land, and which way is an empirical question the data resolve—humidity-adjusted nights *raise* cereal yields, the signature of net augmentation. The distinction has bite only because labour and land are gross substitutes—the farm can trade one for the other, an elasticity of substitution σ above one. A substantial literature on agricultural production rests on this substitutability: the induced-innovation tradition turns on exactly the land–labour margin (Hayami and Ruttan 1985), classic factor-demand estimates find land and labour to be substitutes (Binswanger 1974), and the production-relations literature for Indian (ICRISAT) village agriculture builds on the same substitutability (Binswanger and Rosenzweig 1986), while cross-country CES estimates cluster around unity and vary with method and setting (Mundlak, Butzer, and Larson 2012). I do not estimate σ ; the data discipline it, since the night/day divergence I document can arise only under gross substitutability (the Cobb–Douglas case is observationally silent; online appendix), so the divergence is itself evidence that $\sigma > 1$. When a warmer night changes the quality of an acre, a farm that substitutes readily reworks its land-to-labour mix and works the land harder, hiring *more* field labour rather than less; the factor the shock lands on sets the direction of the labour response, independent of what happens to yields. A neutral productivity loss lands on no factor in particular: it leaves the land-to-labour ratio intact and shrinks the scale of the operation, pulling down the marginal product of labour and the field wage and, where the food market is

tight, raising the harvest price. The two margins therefore move farm-labour demand in *opposite* directions—warmer nights raise it, warmer days lower it. That opposition is what the decomposition turns on, and gross substitution is what makes it possible: under Cobb–Douglas ($\sigma = 1$) a land-augmenting shock folds into total factor productivity and becomes observationally neutral, the two channels collapse, and a daily average loses nothing.

Where the labour goes: the seasonal sink. Workers divide across four states—agricultural wage labour, self-cultivation, non-agricultural work, and casual seasonal work—whose shares sum to one, so the margins move as one offsetting system, and the casual seasonal pool is the flexible residual that clears the reallocation. When nighttime warming raises the demand for farm labour, the farm hires: the wage-labour share rises and draws from the cultivators who give up their marginal plots and the casual workers who attach to the harvest—the mark of a commercialising farm that retains and reorganises its labour. When daytime warming collapses the marginal product, the wage-labour market gives way and returns workers to the seasonal margin. Self-cultivation falls under both margins, as marginal plots go unworked.

The price test of market integration. I test the decomposition’s openness rather than assume it. Only a productivity-driven (daytime) output loss raises the local harvest price; the nighttime contraction leaves it flat. That asymmetry marks a district whose food price is set largely by trade, not by local scarcity (partial market integration). The online appendix embeds this in a general-equilibrium closure spanning open and closed food markets and shows that the over-identifying price restriction places India’s districts on the integrated side; a fully closed subsistence economy would instead invert the labour orientation. Partial integration lets a daytime output loss raise the local harvest price, which is why only the daytime margin moves it.

Rural areas, towns, and the reading of the results. Rural and urban populations share the signs and differ only in amplitude. The factor-bias divergence is a mechanism that operates inside the agricultural sector; its footprint in local employment shares therefore scales directly with that sector’s weight in the local economy. Where agriculture is the dominant sector—rural district populations—the reallocation registers sharply, amplified further by the depth of the casual-labour margin that absorbs it. Where agriculture is a minor sector—towns—the within-agriculture divergence attenuates and the dominant sector’s own heat response shows through: the shared loss of capacity for outdoor informal work contracts non-agricultural employment under both margins—the long-run restraint of non-farm demand under heat (Liu, Shamdasani, and Taraz 2023; Nath 2025)—though, as the wet-bulb cut shows, this attenuation need not be exact.

The framework belongs to the structural-transformation tradition of Matsuyama (1992), Gollin and Rogerson (2014), and Bustos, Caprettini, and Ponticelli (2016); the online appendix develops the full general-equilibrium model—primitives, the gross-substitutes assumption, the price closure, and the lemma and propositions behind the predicted signs, with all proofs. These signs guide the analysis rather than serving as a structure to be estimated; the next sections test them.

2. Data and Empirical Approach

2.1. Data Sources

I harmonise four datasets to link climate, agricultural production and structural transformation: (1) daily gridded weather data from the IMD, (2) district agricultural data (ICRISAT), (3) district and town level tables from the Indian Census, and (4) urban household-level employment data from the NSS. These datasets span four decades (Census, IMD, ICRISAT), with a shorter window from the NSS (1987, 1999, 2009).

Indian Meteorological Department. Gridded data were obtained from the Indian Meteorological Department (IMD) on air temperature (1° resolution) and precipitation (0.25° resolution) over 1951–2023. Consistent with the approach in Liu, Shamdasani, and Taraz (2023), daily observations are averaged over growing seasons of the past 10 years. The growing season marks the months of June–February, which captures cropping cycles of rice in the monsoon (kharif) and wheat in the winter (rabi) seasons. The baseline exposure measures are the growing-season means of the daily minimum and maximum surface-air temperatures, T_{min} and T_{max} , which separate nighttime from daytime warming. For comparison I also construct degree days (DDs), a scalar that rescales and sums agriculturally relevant temperatures between 8 and 32°C (Blakeslee and Fishman 2015; Schlenker, Hanemann, and C. 2006); degree days average over the diurnal cycle and so collapse the two margins into one.

$$(1) \quad DDs = \sum_d D(T_d) ; \text{ where } T_d = \frac{T_{min,d} + T_{max,d}}{2}$$

$$(2) \quad D(T) = \begin{cases} 0, & \text{if } T \leq 8^\circ\text{C} \\ T - 8, & \text{if } 8^\circ\text{C} < T \leq 32^\circ\text{C} \\ 24, & \text{if } T > 32^\circ\text{C} \end{cases}$$

As a humidity-adjusted robustness measure I also construct growing-season wet-bulb minimum and maximum temperatures following Stull (2011); these are derived from

T_{min} and T_{max} alone—no external humidity series is used (the construction is detailed in the supplemental data appendix)—and enter the wet-bulb specifications reported there.

Census of India. The core dataset is sourced from the District Census Handbooks (DCHBs) of India. Specifically, I digitise the Primary Census Abstract (PCA) tables, covering the universe of Indian towns from 1981 to 2011.⁴ A Census town meets three criteria: (1) a minimum population of 4,000, (2) at least 75% of the male working population in non-agricultural activity, and (3) a population density above 400 persons per sq. km. I construct a panel of 2,969 towns that appear consistently across all four decades, so that the analysis reflects intensive-margin adjustments rather than the reclassification of urban areas. For comparability over time, occupations are aggregated into four mutually exclusive categories: cultivators (landowners), agricultural labourers (wage workers), non-agricultural workers, and marginal workers. The “marginal” category cannot be assigned cleanly to agriculture or non-agriculture, but it lets me track the contractual *quality* of work.

ICRISAT. The ICRISAT District Level Database (DLD) tracks agricultural inputs and production in India from 1966 to the present. I harmonise districts to 2011-Census consistent boundaries, constructing an annual panel of cultivated area, production, yields and harvest prices for key grains: rice, wheat, maize, sorghum, barley and millets. I construct the mean of these measures over 10 years preceding each Census round, e.g. over the years 1971–1980 for the 1981 Census. Thus, the results from the temperature regressions compare contemporaneous horizons of 10-year windows with climate data. The DLD covers staple grains only; horticulture and cash crops—which have grown rapidly and differ in labour intensity, irrigation, and market structure—lie outside it, so the agricultural-economy results speak to staple-grain production and the broader composition of Indian agriculture is left to future work. The Census labour reallocation, however, survives this composition shift: it is strongest in districts that stayed in staple grains and muted where agriculture diversified into high-value crops (online appendix).

National Sample Surveys. I use three rounds of the National Sample Survey (NSS) Employment and Unemployment Schedules (1987, 1999, and 2009), harmonised by IPUMS (Ruggles et al. 2024). While the Census measures the volume of labour reallocation, it lacks a granular view of the destination sectors within non-agriculture under its 1981 definitions. Restricting the NSS sample to urban households serves as an independent benchmark to the Census urban and town-level data. The NSS urban sampling frame is separate from Census towns (it retains more peri-urban agriculture), and provides complementary evidence of the divergence in agricultural wage-labour shares. The data

⁴The town digitisation builds on data assembled for Blakeslee et al. (2023).

additionally shed light on the destination of heat-displaced labour—casual, informal work rather than stable formal employment in manufacturing or services.

2.2. Empirical Approach

The identification relies on two approaches: a fixed effects panel estimation (3), and a long differences approach (4) described under Dell, Jones, and Olken (2014), Burke and Emerick (2016) and Liu, Shamdasani, and Taraz (2023). Let Y_{irt} be the outcome of interest (e.g., share of workers in agriculture) for district or town i , within state r , observed at time t . $\mathbf{T}_{it}$ denotes a vector of temperature measures, either degree days $\mathbf{T} = \{DDs\}$ or the diurnal surface-air pair $\mathbf{T} = \{T_{min}, T_{max}\}$. All specifications control for total precipitation $\mathbf{P}_{it}$ and fit a state-specific linear time trend.

$$(3) \quad Y_{irt} = \beta \mathbf{T}_{irt} + \phi \mathbf{P}_{irt} + \mu_i + \lambda_t + \gamma_r \cdot t + \varepsilon_{irt}$$

$$(4) \quad \Delta Y_{ir} = \beta \Delta \mathbf{T}_{ir} + \phi \Delta \mathbf{P}_{ir} + \gamma_r + \varepsilon_{ir}$$

For temporal parity, I construct the gridded temperature and rainfall measures as a moving average over the growing seasons of the preceding ten years. Decadal climate is the inverse-distance-squared-weighted average of IMD grid points surrounding the district (within 200 km) or town (within 100 km) centroid—radii chosen to match the distance at which the residual spatial correlation of the outcomes decays (roughly 150–200 km for districts and 100 km for towns). Because the weights fall off as the squared distance, grid points beyond roughly 100 km contribute negligibly, so the assignment is insensitive to the exact cutoff.

Equation 3 is a district- or town-decade panel with two-way fixed effects and a state-specific linear trend; because the climate regressor is itself a ten-year moving average, I read its coefficients as the *decadal* response of structural transformation to heat. Equation 4 averages the outcome Y and covariates X over two endpoints—1981–91 and 2001–11—and regresses $\Delta Y = Y_2 - Y_1$ on $\Delta X = X_2 - X_1$. This long-differenced estimator recovers the effect of a sustained increase in $\mathbf{T}$ on the outcome trend ΔY , measured against its secular trend $\Delta \bar{Y}$ over the sample. Both estimators use the same decadal smoothed climate; the long difference adds one further decade of horizon and absorbs unobserved state-level trajectories ($\gamma_r \cdot t$). I treat it as a robustness design—confirming that the decadal-panel result survives a different fixed-effects and averaging structure that strips decade-common shocks—rather than as a distinct long-run or adaptation-inclusive

estimand.⁵

I enter T_{min} and T_{max} simultaneously and read each coefficient as the partial effect of one margin holding the other—and rainfall—constant. Inference faces two distinct dependencies. The first is the decadal *persistence* of the climate regressor (a ten-year moving average), which inflates rejection rates unless the estimator accounts for within-unit serial correlation. A placebo Monte Carlo in the online appendix disciplines this—regressing the key outcomes on the real climate series randomly permuted across districts, which preserves each series’ decadal persistence while breaking its link to the outcome—and I select the preferred panel estimator for equation 3 as the specification that rejects the null in roughly 5% of simulations. Unadjusted and heteroskedasticity-robust standard errors over-reject sharply, whereas clustering by district with state-specific linear trends restores the nominal 5% size. The second dependency is spatial; because the permutation reassigns whole series across units it breaks the regressor’s spatial structure, so the Monte Carlo does not speak to spatial correlation, which I address separately with Conley spatial-HAC standard errors (Conley 1999) reported alongside the clustered errors throughout; the online appendix sweeps the Conley radius to 200 km and the headline coefficients remain significant.

The simultaneous estimator addresses two identification challenges. First, because T_{min} and T_{max} are correlated ($\rho \approx 0.78$ across districts), entering either alone would induce omitted-variable bias; entering them together isolates their partial effects and identifies a narrowing versus widening of the local diurnal range. The district and decade fixed effects and state trends absorb most of this co-movement, so the *estimating* variation is close to orthogonal (within-transformed VIF ≈ 1 ; the online appendix) and the partial signs are well-identified. By Frisch–Waugh–Lovell the coefficients load on exactly this residual variation, so the within correlation, rather than the raw ρ , governs whether the two margins can be separated—and within, they are all but orthogonal. Second, absorbing state-specific trends accounts for time-varying state policies and compares districts and towns within their local economic geography under similar baseline climate trajectories. Correlograms of the residuals confirm that the spatial correlation of these heat shocks in the key outcomes decays within roughly 100 km for towns and 150–200 km for districts.

⁵Lemoine (2021) shows that long-difference and panel estimators of climate impacts are identified by overlapping variation in the realised temperature series (here decadal averaged), so a long difference need not isolate differential rates of climate change or a separate long-run adaptation response, and agreement between the two designs reflects robustness to the estimating structure rather than evidence on adaptation dynamics. Identification here therefore rests on the within-unit, simultaneous T_{min} -versus- T_{max} contrast in the decadal panel; the ten-year smoothing and the long difference serve as robustness rather than as a short-run/long-run decomposition.

3. Results

TABLE 1. Panel estimates of diurnal warming on the district agricultural economy

	Log Levels (All Grains)				
	Cultivated Area (1)	Production (2)	Harvest Price Index (3)	Field Wage (4)	Yield Index (5)
T_{min}	-0.218 (-4.76) [-4.76]	-0.145 (-2.26) [-2.25]	-0.042 (-1.36) [-1.36]	-0.038 (-0.62) [-0.63]	0.051 (1.27) [1.28]
T_{max}	-0.225 (-4.01) [-4.02]	-0.221 (-3.47) [-3.46]	0.130 (3.81) [3.77]	-0.119 (-1.70) [-1.76]	-0.046 (-0.97) [-0.97]
Observations	1,243	1,243	1,191	1,115	1,211
Dep Var. Mean	5.45	5.77	5.66	2.86	7.14
F-stat ($T_{min} = T_{max}$)	0.01	0.80	13.34	0.74	2.63
p -value	0.905	0.373	<0.001	0.391	0.106
Rainfall Control	Y	Y	Y	Y	Y
State Linear Trend	Y	Y	Y	Y	Y

Notes: Log cultivated area, log production, a log harvest-price index, the log field wage, and an area-weighted yield index for the major Indian cereals (rice, wheat, barley, maize, millets, sorghum), from the ICRISAT district panel. T_{min} and T_{max} are the growing-season (June–February) surface-air minimum and maximum, averaged over the preceding decade; all specifications control for total precipitation over the same window. The data form a decadal panel (1981–2011) on 2011 Census-consistent district boundaries, with district and decade two-way fixed effects and a state $\times$ decade linear trend. Parentheses report t -statistics from district-clustered standard errors; brackets report t -statistics from Conley spatial-temporal HAC standard errors (50 km, four decadal lags); bold coefficients indicate $|t| > 2$. The F -statistic and its p -value test $T_{min} = T_{max}$.

Tables 1 and 2 decompose the effects of warming nights and days on the district agricultural economy: Table 1 reads down the supply chain for the major cereals—rice, wheat, maize, sorghum, barley, and millets—and Table 2 reports the within-district reallocation of labour. The nighttime minimum and the daytime maximum carry opposite signs across the price and labour outcomes, consistent with the theoretical framework.

Both margins shrink the farm. The contraction in cultivated land is statistically identical across the two margins—a 1°C rise cuts log area by 0.22 either way ($F = 0.01$)—while production falls comparably rather than identically (-0.15 at night, -0.22 by day, $F = 0.8$). I read the milder nighttime loss as labour intensification at work: the significant nighttime rise in wage labour adds output back through the production identity, cushioning the extensive-area loss, while daytime labour-shedding compounds

TABLE 2. Panel estimates of diurnal warming on district labour reallocation

	Share of Workers				Log Workers (5)
	Ag. Labour (1)	Cultivator (2)	Seasonal (3)	Non- Agriculture (4)	
T_{min}	0.044 (4.59) [4.65]	-0.028 (-2.21) [-2.23]	-0.033 (-2.83) [-2.83]	0.017 (1.46) [1.48]	0.175 (1.99) [2.02]
T_{max}	-0.026 (-2.76) [-2.77]	-0.010 (-0.83) [-0.84]	0.032 (2.45) [2.43]	0.004 (0.32) [0.32]	-0.017 (-0.20) [-0.21]
Observations	1,187	1,187	1,187	1,187	1,187
Dep Var. Mean	0.18	0.32	0.17	0.32	13.45
F-stat ($T_{min} = T_{max}$)	24.91	0.85	14.66	0.54	2.36
p -value	<0.001	0.356	<0.001	0.462	0.125
Rainfall Control	Y	Y	Y	Y	Y
State Linear Trend	Y	Y	Y	Y	Y

Notes: District shares of total workers by category—agricultural wage labour, cultivators, the casual seasonal margin, and non-agriculture—and, in the last column, the log number of district workers, from the Census district panel. Outcomes, sample, fixed effects, and inference follow Table 1: a decadal panel (1981–2011) with district and decade fixed effects and a state $\times$ decade linear trend, a precipitation control, district-clustered t -statistics in parentheses over Conley spatial-temporal HAC t -statistics in brackets, and bold coefficients for $|t| > 2$. The F -statistic and its p -value test $T_{min} = T_{max}$.

it. The margins part on the price. The daytime maximum raises the log harvest price by 0.13 while the nighttime minimum holds it flat, a split sharp at $F = 13.34$; the field wage edges down under both (-0.12 by day, -0.04 at night, too close to separate). Daytime warming acts as a supply shock: the daytime loss of land and grain passes straight into a dearer harvest price, while the nighttime price holds flat. Cereal yields move little under either margin in the dry-bulb baseline; the augmentation it implies surfaces only under humidity-adjusted temperatures, where night yields rise outright (online appendix). The aggregate also blends opposing grain-level responses—heat-tolerant coarse cereals gain under warmer nights while wheat loses—so the cereal index is a crop-composition average (online appendix).

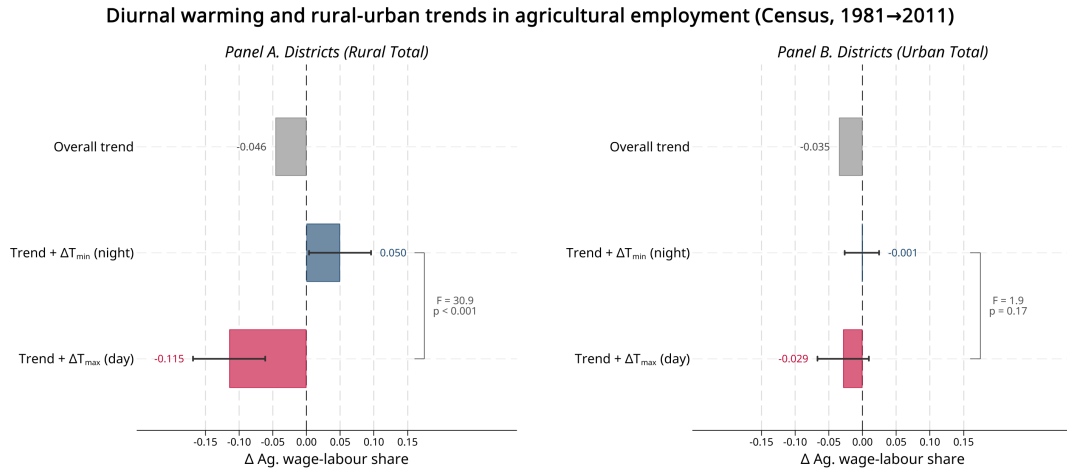
Table 2 locates the labour. A warmer night raises the agricultural wage-labour share by 4.4 points and draws that labour off self-cultivation (-2.8) and the casual seasonal margin (-3.3); a warmer day reverses the flow, cutting the wage-labour share by 2.6 points and returning 3.2 points to the seasonal margin. The two margins separate sharply— $F = 24.9$ on wage labour, $F = 14.7$ on the seasonal share—while non-agricultural employment holds flat under both ($F = 0.5$): the district reshuffles labour within agriculture. The extensive count rises at night, a warmer minimum lifting the log number of district workers by 0.18. The appendix framework grounds the split: nighttime warming acts on land as a land-biased shock, and because labour and land are gross substitutes it raises the demand for field labour directly; daytime warming is a neutral productivity loss that shrinks the operation and sheds labour to the seasonal pool.

Higher nighttimes consolidate agriculture. Though the loss in grain production is comparable to higher daytimes, the district heads off the price shock by intensifying agricultural labour on surviving land. Higher daytimes casualise it. Facing a total productivity loss, the farm contracts in area and production: the wage-labour market gives way, workers fall back to the casual seasonal margin, and the supply loss passes straight through to the harvest-price spike.

Figure 2 carries the labour decomposition to its two-decade horizon and splits each district into its rural and urban populations. The long difference averages each district's two Census endpoints—1981 and 1991 against 2001 and 2011—and differences them, stripping decade-specific noise to isolate the change in climate against the change in labour. The grey bars set the stage: the agricultural wage-labour share fell across rural and urban India over these two decades, the secular structural transformation drawing labour off the land.

Warming bends that trend, and the two margins bend it in opposite directions. A warmer night raises the rural agricultural wage-labour share by 0.096 and turns its -0.046 secular decline into a $+0.050$ rise: nighttime warming hoards labour on the farm against the structural trend. A warmer day lowers the same share by 0.068, deepening

FIGURE 2. Long-differenced estimates of diurnal warming on agricultural labour trends



Notes: Long-difference estimates (1981–2011) of a 1°C rise in growing-season nighttime (T_{min}) and daytime (T_{max}) surface-air temperature on the district agricultural wage-labour share, estimated separately for the rural and urban populations of each district. Grey bars give the secular trend (the mean change in the share); each coloured bar adds the warming coefficient to that trend ($\bar{y} + \hat{\beta}$), with 95% confidence intervals (spikes); the brace reports the night-versus-day equality test ($\Delta T_{min} = \Delta T_{max}$). A coloured bar whose interval clears its grey trend bar is a statistically significant warming effect. All specifications control for the long-run change in precipitation and include state fixed effects; standard errors are clustered at the district level.

the decline to -0.114 : daytime heat pushes a rural agricultural exit. The opposition is sharp and holds over the two-decade horizon.

The urban populations of the same districts barely move. A warmer night lifts the urban agricultural share by 0.034 —a third of the rural response—and a warmer day leaves it flat ($+0.006$, insignificant). The reversal is a rural phenomenon; the urban populations of these districts absorb little of it. The framework reads this as amplitude (online appendix): the diurnal response scales with the agricultural share of local employment, deep in rural districts and thin in urban ones, so the same factor-bias force registers sharply in rural areas and faintly in the town. The online appendix reports the full rural and urban decomposition behind the figure—all four worker shares and log total workers. District populations move in the same direction as workers, with the urban population rising under nighttime warming (online appendix).

TABLE 3. Long-differenced estimates of diurnal warming on labour reallocation in towns

	Share of Workers				Log Total Workers
	Ag. Labour (1)	Cultivator (2)	Seasonal (3)	Non-Agriculture (4)	
ΔT_{min}	0.063 (5.46)	-0.009 (-0.86)	-0.016 (-1.35)	-0.038 (-2.16)	0.272 (1.61)
ΔT_{max}	0.041 (2.59)	0.022 (1.49)	0.017 (1.15)	-0.078 (-3.47)	0.241 (1.41)
Observations	2,969	2,969	2,969	2,969	2,969
F-stat ($\Delta T_{min} = \Delta T_{max}$)	1.61	3.38	4.06	2.78	0.11
<i>p</i> -value	0.205	0.067	0.044	0.096	0.738
Rainfall Control	Y	Y	Y	Y	Y
FE / Trend	Y	Y	Y	Y	Y

Notes: Census-town worker reallocation under nighttime (ΔT_{min}) and daytime (ΔT_{max}) growing-season (June–February) surface-air warming, across the balanced panel of Census towns. The specification is the single long difference (1981–2011) of each share on the long-run change in temperature, with state fixed effects and a precipitation control. The four mutually exclusive Census worker categories—agricultural wage labour, cultivators, seasonal, and non-agricultural—sum to one, so the share coefficients sum to approximately zero; column 5, set apart by a gap, reports the long-run change in log total town workers. The *F*-statistic and its *p*-value test $\Delta T_{min} = \Delta T_{max}$. Parentheses report *t*-statistics from district-clustered standard errors; bold coefficients indicate $|t| > 2$. Long differences eliminate serial correlation; spatial robustness checks are available in the online appendix.

Table 3 carries the worker-share decomposition to the Census town over the long difference (1981–2011). The framework’s prediction for towns is a common drag—where agriculture is too small to register the within-agriculture reallocation, the heat-capacity loss on informal outdoor work depresses the non-agricultural share under *both* margins (online appendix). The long difference bears this out: a warmer night and a warmer day

both cut the town non-agricultural share (-0.038 and -0.078) and both raise agricultural wage labour ($+0.063$, $+0.041$), with little day/night divergence ($F = 1.6$ on wage labour). Both ends of the diurnal cycle hold labour out of town non-agriculture. This town de-urbanisation mirrors Blakeslee et al. (2023), where irrigation likewise raises rural agriculture and contracts the town non-farm sector (online appendix).

A decadal town panel, reported in the online appendix, shows the same non-agricultural margins muted towards zero alongside a residual night-warming pull onto town agriculture—a warmer night raises the town wage-labour share by 1.7 points ($F = 7.2$ against a flat daytime coefficient) and lifts the town workforce ($+0.19$). I read the common non-agricultural drag, present in both the panel and the long difference, as the town counterpart of the district result; the smaller nighttime pull is a less robust signal that the log-linear framework treats town agriculture as too thin to generate. The balanced-versus-statutory-town decomposition, including the in-migration that lets established towns grow their workforce even as the non-farm share contracts, is in the online appendix.

A final, suggestive question is where the displaced labour lands. Matched to three rounds of the National Sample Survey, the urban data reproduce the district mechanism on its matched margin—a warmer night raises the urban agricultural wage-labour share ($+0.060$, $p < 0.01$)—and hint that daytime warming diverts labour into informal trade and transport rather than formal employment (online appendix). With only three survey rounds I read this as suggestive and leave it to future work.

The online appendix confirms the decomposition under alternative measures: a single degree-days measure collapses the opposition into one muted coefficient; and wet-bulb temperatures sharpen the labour split while humid nights raise cereal yields outright ($+0.15$), the land-augmenting signature ($B' > 0$) absent from the dry-bulb baseline, where the night margin is on net contractionary.

4. Conclusion

This paper asks which way warming temperatures move agricultural labour. To my knowledge, this paper provides the first reduced-form evidence and a conceptual framework for understanding the economic channels of diurnal warming — shifts in nighttime and daytime dry-bulb air temperatures — on structural transformation. The framework carries a single over-identifying restriction: only the daytime margin moves local harvest prices. The data confirms this asymmetry ($F = 13.34$ on equality of $\beta_{T_{\min}} = \beta_{T_{\max}}$ on prices), providing an empirical grounding for partial market integration — independently of the labour reallocation. The framework is a guide on the signs rather than magnitudes, and the empirical approach of a panel and long-differenced estimation tests the direction on key comparative statics.

Though the diurnal divergence is robust to a battery of stress tests, the empirical approach has several important nuances and limitations. First, while multicollinearity of the within-variation (ρ) is a concern for identification under wet-bulb temperatures $\rho \approx 0.84$, in the surface-air baseline the two margins are within-correlated at only $\rho \approx -0.09$ (online appendix). Twoway fixed effects and state trends absorb almost all of that co-movement, leaving the identifying variation near-orthogonal (within-transformed VIF ≈ 1 , condition number ≈ 1.1). Both $T_{\min}$, $T_{\max}$ carry their sign on agricultural labour shares (+/-, respectively) when specified independently with and without rainfall controls in both panel and long differenced estimations. The within-correlation widens standard errors without drastically altering the magnitude and sign of the point estimates $\beta_{T_{\min}}, \beta_{T_{\max}}$.

Second, the wet-bulb temperature specification is a deterministic nonlinear approximation using the raw dry-bulb $T_{\min}$ and $T_{\max}$ (online appendix), mechanically compressing the diurnal temperature range (DTR). This makes the wet-bulb pair highly collinear relative to the dry-bulb baseline (within- $\rho \approx 0.84$, within-transformed VIF ≈ 3.5 ; the online appendix).

Third, specified jointly, $T_{\min}$ and $T_{\max}$ span the same two-dimensional regressor space as the daily mean and the diurnal range (DTR), so the diurnal decomposition mirrors temperature level vs range effects, uninfluenced by the chosen basis. A mean-range reparametrisation provides supporting evidence that the DTR carries an independent channel on the labour and price margins (online appendix).

Finally, the divergence further assumes labour and land are gross substitutes ($\sigma > 1$) in the conceptual framework. I do not directly estimate this assumption, though it is grounded in the agricultural-production literature (Hayami and Ruttan 1985; Binswanger 1974). The data disciplines this assumption, as the night/day divergence can arise only under gross substitutability (online appendix). Nonlinearity and spatial sorting, held fixed by the linear framework, are left for future research.

A natural extension of the framework's distinctive prediction is through India's own pre-liberalisation Censuses and colonial agricultural statistics, which provide a test for whether a fully closed subsistence economy without buffers inverts the night-day pattern. The pattern itself is consistent: warmer nights hold labour on the farm without the price and wage distress that warmer days bring, while daytime heat restrains the non-agricultural transition far more strongly. Given the two margins carry opposite signs on agricultural labour, the trajectory of the diurnal temperature range matters as much as the level. Where nights are warming faster than days—compressing the range—the decomposition implies agricultural consolidation and commercialisation, while where days outpace nights and the range widens it implies the double squeeze of farm-labour casualisation and stalled urban structural transformation.

A back-of-the-envelope calculation suggests over 1981–2011 growing-season nights and days warmed by 0.45° and 0.37°C on average, but the balance is regional—in the districts where the range compressed, nights ran ahead of days ($+0.71$ versus $+0.17^{\circ}\text{C}$), and where it widened, days led ($+0.54$ versus $+0.21^{\circ}\text{C}$). Applying panel estimates from Table 2, the range-compressing $T_{\min}$ warming implies a 2.7-point rise in the agricultural wage-labour share against an 18% base. The range-widening $T_{\max}$ warming implies a slight decline and a 1.0-point shift onto the casual seasonal margin. Night and day warming, in equivalent magnitudes, sort local agricultural labour in opposite directions. Under extreme daytime heat, the exit from agricultural labour into seasonal, informal employment reflects a physiologically constrained churn for survival, rather than a sustained pathway to structural change.

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